

CHEE HENG KOH

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EDUCATION

National University of Singapore (NUS)

Jul 2021 - May 2025

Bachelor of Engineering (Computer Engineering)

- GPA: 4.74/5.00, Honours (Highest Distinction); Dean's List AY2022/23
- Relevant Courses: Data Structures and Algorithms, Software Engineering, Computer Networks, Operating Systems, Database Management

CORE COMPETENCIES

- Programming Languages:** Java, Go, Python, JavaScript, C++
- Tech Stack:** Apache Kafka, gRPC, Redis, Elasticsearch, MySQL, MongoDB, Docker, Git

WORK EXPERIENCE

Backend Software Engineer, ByteDance / TikTok, Singapore

Jul 2025 – Present

- Led end-to-end development, evolution, and performance optimisation of a unified Workforce & Operations Platform, building an agent-facing portal and shared services architecture supporting ~1,000 DAU, 4,000+ daily requests, and 700+ active workflows at >99.9% availability.
- Owned core workforce management capabilities spanning forecasting, scheduling, compliance approval workflows, and dynamic rule-based group membership infrastructure powering 265+ active groups, enabling operational coordination across the team's services.
- Designed and delivered shared backend infrastructure adopted across the team's operations management services, including a Redis-backed distributed lock service used by ~70% of services and a centralised serverless job orchestration system that migrated all scheduled workloads, significantly reducing integration overhead and onboarding effort.
- Led system reliability and disaster recovery initiatives across multiple services, designing service-level multi-datacenter failover strategies over Redis, Elasticsearch, database, and messaging dependencies. Achieved ≤15-minute failover capability, reduced critical endpoint latencies by up to 66.6%, and resolved long-standing data correctness issues.

Full-Stack Engineer Intern, HubbedIn, Singapore

Jul 2024 – Nov 2024

- Architected and developed microservices for an Enterprise Resource Planning system using Node.js and RabbitMQ, enabling real-time synchronisation across services and expanding the product into physical retail environments, targeting deployment across eight stores.
- Designed and delivered a financial accounting module with MongoDB, enabling accurate monthly reconciliation for B2B sales and point-of-sale transactions.

PROJECTS

Full-Stack Developer, Inventory Management System, Singapore

Apr 2023 – Jan 2025

- Constructed an inventory management system using ReactJS and Django for a non-profit first aid organisation, optimising logistics management and reducing incidents of logistical errors during field deployments.
- Awarded the NUS Public Service Fellowship for the project, in recognition of its impact and contributions to the field.

EXTRA CURRICULAR / OTHER ACTIVITIES

Top 10 Finalist, SRA Retail Reimagined Innovation Challenge, Singapore

May 2022 - Jul 2022

- Proposed Augmented Reality and Blockchain solution to help retail businesses adopt Web3.0 technologies.
- Clinched the Finalist and Best Teamwork awards with a diverse team comprising members from different universities.